

$$\begin{aligned} & \frac{1}{6\pi^2} \left(\frac{1}{4} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{2,2,0}[m_1, \tilde{\mu}] + \frac{1}{12} m_2 \tilde{\mu} g_2^4 (9 y_e^{i1i2} - 4 y_e^{pi2} \delta_{pi1}) \text{LF}_{2,2,0}[m_2, \tilde{\mu}] + \right. \\ & \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_e^{pi2} \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] \delta_{pi1} + \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \\ & \text{LF}_{2,2,0}[m_d^r, m_q^p] - \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[m_d^r, m_q^p] + \\ & \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{2,2,0}[m_e^r, m_l^p] - \\ & \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[m_e^r, m_l^p] - \\ & \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,1,0}[m_l^p, m_e^r] - \\ & \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[m_l^p, m_e^r] + \\ & \frac{1}{8} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} (3 g_1^2 - g_2^2) (y_e^{i1i2} + y_e^{si2} \delta_{si1}) - 6 g_1^2 y_e^{i1s} \delta_{si2}) \text{LF}_{4,1,-1}[m_l^p, m_e^r] + \\ & \frac{1}{6} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} (g_2^2 y_e^{si2} \delta_{si1} - g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[m_l^p, m_e^r] - \\ & \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,1,0}[m_q^p, m_d^r] - \\ & \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[m_q^p, m_d^r] + \\ & \frac{1}{6} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} (5 g_1^2 - 3 g_2^2) (y_e^{i1i2} + y_e^{si2} \delta_{si1}) - 10 g_1^2 y_e^{i1s} \delta_{si2}) \text{LF}_{4,1,-1}[m_q^p, m_d^r] + \\ & \frac{1}{2} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} (g_2^2 y_e^{si2} \delta_{si1} - g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[m_q^p, m_d^r] - \\ & \frac{1}{12} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (3 g_2^2 y_e^{si2} \delta_{si1} + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{2,2,0}[m_q^p, m_u^r] + \\ & \frac{3}{4} \tilde{\mu} g_2^2 y_e^{i1i2} \overline{y_u^{pr}} a_u^{pr} \text{LF}_{3,1,0}[m_q^p, m_u^r] + \\ & \frac{1}{24} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (3 g_2^2 (-3 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\ & \text{LF}_{3,2,-1}[m_q^p, m_u^r] - \frac{3}{4} \tilde{\mu} g_2^2 y_e^{i1i2} \overline{y_u^{pr}} a_u^{pr} \text{LF}_{4,1,-1}[m_q^p, m_u^r] + \\ & \frac{1}{12} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (3 g_2^2 y_e^{si2} \delta_{si1} + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{3,1,0}[m_u^r, m_q^p] + \\ & \frac{1}{12} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (3 g_2^2 y_e^{si2} \delta_{si1} + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{3,2,-1}[m_u^r, m_q^p] + \\ & \frac{1}{4} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (-3 g_2^2 y_e^{si2} \delta_{si1} + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{4,1,-1}[m_u^r, m_q^p] + \\ & \frac{1}{2} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (g_2^2 y_e^{si2} \delta_{si1} - g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[m_u^r, m_q^p] + \\ & \frac{5}{8} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\ & \frac{1}{8} m_1 \tilde{\mu} g_1^2 (-g_2^2 y_e^{pi2} \delta_{pi1} + g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] - \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \\ & \frac{1}{6} m_1 \tilde{\mu} g_1^2 (g_2^2 y_e^{pi2} \delta_{pi1} - g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \\ & \frac{1}{3} m_2 \tilde{\mu} g_2^4 y_e^{pi2} \text{LF}_{3,1,0}[\tilde{\mu}, m_2] \delta_{pi1} + \frac{1}{24} m_2 \tilde{\mu} g_2^4 (51 y_e^{i1i2} + 8 y_e^{pi2} \delta_{pi1}) \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\ & \frac{1}{8} m_2 \tilde{\mu} g_2^2 (-7 g_2^2 y_e^{pi2} \delta_{pi1} + 3 g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] - \\ & 2 m_2 \tilde{\mu} g_2^4 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\ & \frac{1}{2} m_2 \tilde{\mu} g_2^2 (g_2^2 y_e^{pi2} \delta_{pi1} - g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] - \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}[m_1, m_e^r, m_l^p] + \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}[m_1, m_e^r, m_l^p] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \\ & \text{LF}_{2,1,1,0}[m_1, m_e^r, \tilde{\mu}] - \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}[m_1, m_e^r, \tilde{\mu}] - \\ & m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{3,1,1,-1}[m_1, m_e^{i2}, \tilde{\mu}] - \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}[m_1, m_l^p, \tilde{\mu}] + \\ & \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}[m_1, m_l^p, \tilde{\mu}] - \\ & \frac{1}{4} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{3,1,1,-1}[m_1, m_l^{i1}, \tilde{\mu}] - \frac{5}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^r] - \\ & \frac{3}{2} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^{i2}] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^p] + \\ & \frac{3}{8} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{i1}] + \frac{3}{4} m_2 \tilde{\mu} g_2^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}[m_2, m_l^p, \tilde{\mu}] - \\ & \frac{3}{4} m_2 \tilde{\mu} g_2^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}[m_2, m_l^p, \tilde{\mu}] - \\ & \frac{3}{4} m_2 \tilde{\mu} g_2^4 y_e^{i1i2} \text{LF}_{3,1,1,-1}[m_2, m_l^{i1}, \tilde{\mu}] + \frac{3}{2} \tilde{\mu} g_1^2 g_2^2 y_e^{i1i2} (m_1 + m_2) \text{LF}_{2,2,1,-1$$